CCINFOM - S22 - Group 3

Database Application Proposal

Submitted by:

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Submitted to:

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Section 1.0 - Group Composition

1. GONZALES, Aaron James S.
2. MANALANG, Kennese Ross F.
3. MARCAIDA, Duncan Joseph B.
4. PACOMIO, Leelancze Naethan N.

Section 2.0 - Why is this Database System important to be developed?

In logistics companies, clients, vehicles, drivers, and vehicle parts must be tracked accurately to ensure smooth operations. A centralized database system provides a structured and timely way of managing records, preventing data loss and errors that could occur with manual tracking methods such as Excel sheets. Through this service, businesses can efficiently track the status of vehicle conditions and part availability, completion of shipment, and documentation of incidents for transparency and operational efficiency.

Section 3.0 - Records Management

Records	**Fields//Outputs**	**Assigned Member**
Vehicle	(vehicle_id, plate_number, vehicle_type, model/Brand, status, fuel_type, mileage) Viewing a Record with Other Related Records: Viewing a vehicle with all the list of drivers that have used it Viewing a vehicle and clients that have associated with it	GONZALES, Aaron James S.
Client	(client_id, client_type, name, contact_person, phone, email, address, priority_flag, status, completed_orders) Viewing a Record with Other Related Records: View a specific client and list of vehicles that have served that client	MANALANG, Kennese Ross F.
Parts	(part_id, part_name, description, stock_qty, cost, supplier, pending_delivery) Viewing a Record with Other Related Records: View a specific part and list of vehicles that require this part	MARCAIDA, Duncan Joseph B.
Driver	(driver_id, firstname, lastname, license_number, contact, email, status, completed_trips) Viewing a Record with Other Related Records: Viewing a specific driver and the list of vehicles they have used.	PACOMIO, Leelancze Naethan N.

Table 1. Core Database Entities

Section 4.0 - Transactions

Transaction	Operations	Involved Records	Assigned Member
Log Fuel	Fuel log(fuel_id, vehicle_id (FK), driver_id (FK), fuel_date, fuel_type, liters_filled, price_per_liter, reimbursed) ● Verify record of the vehicle for current status and mileage before refueling ● Verify record of the driver performing the refueling ● Record details of the fuel purchase (date, time, fuel type, quantity, cost) ● Update vehicle record fuel entry for consumption tracking ● Record driver reference for and trip cost	Vehicle Driver	GONZALES, Aaron James S.
Log Trip/Shipment	trip log(trip_id, client_id (FK), driver_id (FK), vehicle_id (FK), pick_up_loc, drop_off_loc, date_time_start, date_time_completed, trip_cost, status) ● Verify if client is eligible to request shipments ● Verify if vehicle status is available ● Verify if driver is available ● Recording the trip details(pickup, dropoff, etc.) ● Record trip completion details (arrival time, final status, etc.) ● Update Vehicle and Driver Status ● Update vehicle record to add mileage and change status ● Update driver record to increment number of completed trips ● Update client record to increment number of completed orders	Vehicle Client Driver	MANALANG, Kennese Ross F.
Log Maintenance	maintenance log(maintenance_id, vehicle_id (FK), date_time_start,	Vehicle Parts	MARCAIDA, Duncan Joseph B.

	date_time_completed, description, status) ● Verify record of vehicle for current status ● Verify if quantity of stock of parts is enough to perform maintenance ● Record parts consumed during maintenance ● Update status of vehicle to available once completed		
Log Incident	incident_log(incident_id, driver_id (FK), vehicle_id (FK), incident_type, incident_date_time, incident_location, incident_severity) ● Verify record of the driver involved ● Verify record of the vehicle involved ● Record incident details (type, date, time, location, severity) ● Update vehicle status if needed (e.g if the incident caused damage to the vehicle) ● Update driver record if needed (e.g if the driver needs to be suspended)	Driver Vehicle	PACOMIO, Leelancze Naethan N.

Table 2. List of Transactions

Section 5.0 - Reports to be Generated

Report	**Description**	**Time**	**Assigned Member**
Mileage Report	Shows the average mileage per vehicle, per model, within a selected time period.	Per day, for a given Year and Month	GONZALES, Aaron James S.
Driver Performance Report	Shows the number of completed trips, delays, or incidents per driver within a time period	Per week, for a given Year and Month	MANALANG, Kennese Ross F.
Maintenance Cost Report	Shows a summary of maintenance done on a vehicle, as well as the expenses spent, including parts, service costs, and labor, to	For a given year	MARCAIDA, Duncan Joseph B.

	manage and carefully audit expenditures.		
Client Activity Report	Summarizes per client activity such as the number of shipments booked and completed, and vehicles used for that client	Per month, for a given year	PACOMIO, Leelancze Naethan N.

Table 3. List of Generated Reports

Section 6.0 - Declaration of Generative AI Use

We, Aaron James S. Gonzales, Duncan Joseph B. Marcaida, Kennese Ross F. Manalang, Leelancze Naethan N. Pacomio of section S22, declare that no generative AI tools were used in the preparation of this project proposal. All ideas and content are our own work and effort. We take full responsibility for the submission and understand the repercussions of committing academic dishonesty, as stated in the DLSU Student Handbook. We affirm that we have not used any unauthorized assistance or unfair means in completing this project.

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